

Voice Judge-QA — Final Handover

All credentials, endpoints, and verified connectivity for the third-party developer

Issued 2026-05-14 · BizFinder.ai

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1. Credentials

Everything the developer needs to start work. Three credentials, one shared secret.

SSH / SFTP login

HOST	38.247.189.143
PORT	2203
USERNAME	analytics
PASSWORD	h9a0VZB33YZequrS

LOGIN COMMAND	ssh analytics@38.247.189.143 -p 2203
SFTP COMMAND	sftp -P 2203 analytics@38.247.189.143

QA Read API

PRODUCTION BASE URL	https://bizfinder.ai
STAGING BASE URL	https://staging.bizfinder.ai
AUTH HEADER	X-QA-Secret: <secret below>
QA_SHARED_SECRET	fea77692e1d33bc18807e6dda9be4f95db471ba289af20ea737e5009777a47a0

Internal service endpoints

Bot service health URL and the LLM / STT / TTS internal endpoints will be handed off per engagement when the developer needs them for bot-side work. They are not required to use the QA Read API.

Do not commit credentials to source control. Store the QA secret in `~/.qa.env` with `chmod 600` and source it from your scripts.

2. Provisioning Status

Final check ran on 2026-05-14 from inside the `analytics` account against the deployed `bizfinder.ai` production environment.

ITEM	STATUS	DETAIL
Dedicated VPS account (analytics)	DONE	Account active on port 2203, UID 1003, home /home/analytics writable.
SSH access	DONE	Password login confirmed on port 2203. Pre-created subdirs ~/turns/ and ~/recordings/ writable.
Outbound network from the account	DONE	Public internet reachable (verified via curl to bizfinder.ai + TLS chain).
Main app reachable from the account	DONE	bizfinder.ai responds to all three QA endpoints with valid TLS.
QA Read API routes	LIVE	/api/qa/health , /api/qa/conversations , /api/qa/conversations/<id-or-sessionId> . Deployed via Coolify, redeploy rtv263kxhgqefq3j3ley9598 finished 21:58 UTC.
QA_SHARED_SECRET on production	SET	Confirmed live on bizfinder.ai .
QA_SHARED_SECRET on staging	SET	Confirmed live on staging.bizfinder.ai .
Dedicated qa-judge siteld	PENDING	BizFinder admin to create a preview business with siteld qa-judge . Suggested before first dev call.
Bot-side audio capture (frame interceptor)	DEV SCOPE	Third-party developer extends bot/app.py . Hook points exist; see section 7.
GPU slot on A6000 for the judge model	DEFERRED	Allocate once the developer is ready for live-call grading. Ollama with OLLAMA_KEEP_ALIVE=5m handles auto-load / auto-unload.

3. SSH / SFTP Access

FIELD	VALUE
Host	38.247.189.143
Port	2203
Username	analytics
Password	h9a0VZB33YZequrS
UID / GID inside the account	1003 / 1003
Resources	2 vCPU, 4 GB RAM
Home directory	/home/analytics
Pre-created subdirs	~/turns/ and ~/recordings/ (writable)
Network	Public internet reachable.

```
# SSH login
ssh analytics@38.247.189.143 -p 2203
# password: h9a0VZB33YZequrS

# SFTP for bulk file transfer
sftp -P 2203 analytics@38.247.189.143

# Once inside, the standard layout:
~/turns/          # per-turn user.mp3 / assistant.mp3 (one dir per sessionId)
~/recordings/     # full-call mp3 (one file per sessionId)
~/.qa.env         # suggested home for the QA secret (chmod 600)
```

4. QA Read API

FIELD	VALUE
Prod base URL	https://bizfinder.ai
Staging base URL	https://staging.bizfinder.ai
Auth header	X-QA-Secret: <value below>
Scope	Read-only. isVoice=true conversations only.
Rate limits	None enforced. Please poll responsibly (≤ 1 req/sec).

```
QA_SHARED_SECRET = fea77692e1d33bc18807e6dda9be4f95db471ba289af20ea737e5009777a47a0
```

5. Live Connectivity Verification

The following matrix was executed on 2026-05-14, signed in as analytics@38.247.189.143:2203 , against the **deployed production** environment. Every check is currently green.

TEST	COMMAND	RESULT	VERDICT
SSH login succeeds	ssh analytics@... -p 2203	Authenticated, password accepted	PASS
Outbound HTTPS reachable	curl https://bizfinder.ai/	HTTP 200, 1.5s, TLS valid	PASS
/api/qa/health with valid secret	curl -H "X-QA-Secret: \$SECRET" .../api/qa/health	HTTP 200, {"ok":true,"service":"qa-read-api","serverTime":"2026-05-14T00:12:34.381Z"}	PASS
/api/qa/conversations?limit=1	+ valid secret	HTTP 200, returns a real voice conversation row with sessionId , siteId , metrics , timestamps	PASS
/api/qa/conversations/<sessionId>	+ valid secret	HTTP 200, returns conversation + 4 ordered messages	PASS
Auth gate — wrong secret	-H "X-QA-Secret: wrong"	HTTP 401, {"error":"Unauthorized"}	PASS
Auth gate — missing header	no header	HTTP 401, {"error":"Unauthorized"}	PASS
SFTP write into ~/turns/	SFTP mkdir + put hello.txt	File created, owner analytics:analytics	PASS
SFTP round-trip into ~/recordings/	4 KB binary put then get	SHA-256 matches end-to-end — binary integrity preserved	PASS

All nine checks green. The QA API is live, the auth gate is enforced, the analytics account can SSH in, read the API, and write audio files back via SFTP into both target directories.

6. Item 2 — Voice Bot Test Endpoint

Where the test caller dials in. Two endpoints exist today; the dev only ever calls the first.

ENDPOINT	METHOD	PURPOSE
POST /api/widget/call (app/api/widget/call/route.ts:21)	POST	Body {siteId, micGranted:true} . Provisions an ephemeral Daily.co room (15-min expiry) and tells the bot service to dial in. Returns {roomUrl, roomToken, botToken} . Public, CORS-open.
POST \$MODAL_START_URL/start (bot/app.py:2038)	POST	Internal — called server-side by /api/widget/call . Spawns the bot thread that joins the Daily room. Dev should not call directly.
GET \$MODAL_START_URL/health (bot/app.py:2100)	GET	Returns {status, models_ready, active_bots, available_slots, threads} . Test caller should refuse to dial when available_slots === 0 .

Bot service URL: handed off separately per engagement.

```
# Test caller flow
curl -X POST https://bizfinder.ai/api/widget/call \
  -H "Content-Type: application/json" \
  -d '{"siteId":"qa-judge","micGranted":true}'

# Returns:
{
  "roomUrl": "https://bizfinder.daily.co/room-abc123",
  "roomToken": "eyJ...",
  "botToken": "eyJ..."
}

# Then join the room using daily-python or a headless browser.
```

7. Item 3 — STT Layer

Catches speech-recognition errors. STT runs **inside** the bot service via Pipecat (not via /api/transcribe , which serves the widget popup path only).

ASPECT	CURRENT STATE
STT provider	LIVE Configurable via <code>stt.provider</code> . Currently a self-hosted Whisper backend, model <code>large-v3-turbo</code> . Provider config in <code>lib/providers/voice.ts:76</code> .
STT output text per turn	LIVE Emitted as <code>TranscriptionFrame</code> at <code>bot/app.py:594</code> . Persisted as <code>Message{role:'user'}</code> rows when the bot POSTs to <code>/api/widget/voice-lead</code> at end-of-call. Accessible via <code>GET /api/qa/conversations/<id></code> .
STT output latency per turn	LIVE <code>Conversation.metrics[].stt_ms</code> , populated by <code>TurnLatencyLogger</code> at <code>bot/app.py:540</code> . Returned by the QA API.
STT input audio (user PCM)	DEV SCOPE Lives only as Pipecat in-memory frames. Not persisted today. Hook points already exist for the dev to extend: <code>UserStartedSpeakingFrame</code> at <code>bot/app.py:480</code> and <code>UserStoppedSpeakingFrame</code> at <code>bot/app.py:488</code> . Capture is gated by a QA flag so non-QA calls remain in-memory-only.

How the judge will score STT: compare the audio file at `~/turns/<sessionId>/<n>-user.mp3` (once the dev wires capture) with `messages[].content` where `role='user'` at the same turn index. Diff = STT errors.

8. Item 4 — LLM Layer

Catches reasoning and hallucination errors. The LLM runs inside the bot service.

ASPECT	CURRENT STATE
LLM call site	LIVE bot/app.py:1001 — OLLamaLLMService against a self-hosted Ollama backend, model qwen2.5:32b . Endpoint URL handed off per engagement.
System prompt	LIVE Fetched from VOICE_PROMPT_URL with a TTL cache (default 300s). Stable per call within the TTL window — dev fetches once and treats as constant. URL to be provided in the per-engagement credentials handoff.
LLM output text per turn	LIVE Message{role: 'assistant'} rows on the Conversation . Accessible via QA API.
LLM latency per turn	LIVE metrics[].llm_ttfb_ms (time to first byte) and metrics[].llm_end_ms (end of generation).
LLM input prompt (full)	RECONSTRUCT Not stored verbatim. Reconstructible by concatenating the system prompt (from VOICE_PROMPT_URL) + the prior messages[] array in chronological order. The QA API returns messages in createdAt asc order to make this trivial.
Raw API trace (request/response)	DEV OPT If the dev needs the raw Ollama JSON traffic, they can mirror it from bot/app.py into the same SFTP destination. Not on the BizFinder roadmap.

How the judge will score LLM: reconstruct {system_prompt, user_turns[]} from the QA API + VOICE_PROMPT_URL , compare with the actual assistant message. Judge the response on relevance, factual grounding (against the per-business KnowledgeChunk rows), and instruction adherence.

9. Item 5 — TTS Layer

The pronunciation layer. This is where the "Rakesh" mispronunciation case surfaces, and the layer that most needs audio capture.

ASPECT	CURRENT STATE
TTS service	LIVE KokoroTTSService subclass at bot/app.py:957. POSTs to tts.kokoro.base_url + /v1/audio/speech . Voice ID per business (Business.voiceId , default af_hear t).
TTS input text	LIVE Same as LLM output. Available in Message{role:'assistant'}.content via QA API.
TTS latency	LIVE metrics[].tts_ttfb_ms — time from text-in to first audio byte out.
TTS output audio (synthesised PCM)	DEV SCOPE Streamed as TTSAudioRawFrame at bot/app.py:977 into the Daily transport. Never written to disk today. Hook points the dev extends: TTSStartedFrame , TTSAudioRawFrame , TTSStoppedFrame .

How the judge will score TTS: the dev's frame interceptor writes the synthesised audio to ~/turns/<sessionId>/<n>-assistant.mp3 . Judge model (e.g. Qwen2.5-Omni or Qwen2-Audio) listens to the file and compares against the messages[].content text. Mismatches like "Ra-keesh" vs the text "Rakesh" surface as TTS pronunciation errors.

10. Item 6 — Call Recording (Full MP3)

Status: DEV SCOPE — not present today. Daily.co cloud recording is *not* enabled on rooms (the room provisioning in app/api/widget/call/route.ts:84 has no enable_recording property, and rooms expire after 15 min). **And we don't want blanket recording** — only the calls the QA team flags.

Chosen approach: bot-side WAV per QA-flagged call. The same frame interceptor the dev builds for items 3 and 5 also writes a full-call WAV concurrently and mp3-encodes on hangup. One code path covers per-turn audio (items 3/5) and full-call audio (item 6).

ASPECT	WHERE IT LANDS
Per-turn audio	~/turns/<sessionId>/<n>-user.mp3 ~/turns/<sessionId>/<n>-assistant.mp3
Full-call audio	~/recordings/<sessionId>.mp3
Transport	Bot host SFTPs to analytics@38.247.189.143:2203
Retention	Dev's choice. Suggested: find ~/turns ~/recordings -mtime +30 -delete via cron under the analytics user.

Hard rule: no QA flag ⇒ no buffer ⇒ no disk write ⇒ no SFTP. Real customer calls stay in-memory-only, exactly as today.

11. Item 7 — Call Logs API with Timestamps

LIVE via the QA API — this entire row is what `/api/qa/conversations*` exposes. Three Prisma models back the API:

MODEL	KEY FIELDS
Conversation (prisma/schema.prisma:273)	sessionId , businessId , isVoice , metrics (JSON), satisfaction , score , createdAt , updatedAt
Message (prisma/schema.prisma:294)	conversationId , role , content , rating , createdAt
DebugLog (prisma/schema.prisma:333)	route , level , message , metadata , durationMs , sessionId , createdAt . Captures route-level events; not currently exposed via QA API, can be added if needed.

METRICS JSON SHAPE (PER TURN)

```
[
  {
    "turn": 0,
    "stt_ms": 420.3,
    "llm_ttfb_ms": 180.7,
    "llm_end_ms": 940.2,
    "tts_ttfb_ms": 95.1,
    "total_ms": 1240.8,
    "user_stopped_at": 1715792134.12,
    "llm_token_count": 38
  },
  ...
]
```

How the judge correlates: use `sessionId` as the join key. Each conversation returns ordered messages with per-message `createdAt` and a parallel `metrics[]` array with timing breakdowns. Pair index `n` across both. Audio files (once captured by the dev) are keyed by the same `sessionId` .

12. Item 8 — GPU Slot on A6000

Status: **DEFERRED** — allocate when the dev is ready for live-call testing. Not a code question; coordinate with BizFinder ops.

The A6000 currently hosts the production STT, LLM, TTS, and embedding models. Total VRAM 48 GB; headroom depends on what's resident at allocation time. Qwen2-Audio-7B fits in ~14 GB.

AUTO-LOAD / AUTO-UNLOAD (RECOMMENDED)

The judge model should not occupy VRAM when no calls are being graded. Three approaches:

RUNTIME	AUTO-UNLOAD?	HOW
Ollama	Yes, native	Set <code>OLLAMA_KEEP_ALIVE=5m</code> on the Ollama process (or pass <code>keep_alive: "5m"</code> per request). Model loads on first request, auto-unloads after 5 min idle. Same Ollama already serves <code>qwen2.5:32b</code> — no new infra. Recommended if a suitable audio-capable model is available in Ollama's library (e.g. <code>qwen2.5-omni</code>).
vLLM	Manual	Run vLLM behind a wrapper that starts the server on first request and shuts it down after idle timeout. Heavier; only worth it if Ollama doesn't carry the chosen model.
Transformers (HF)	Manual	Load on first request, <code>torch.cuda.empty_cache()</code> + <code>del model</code> after idle. Dev owns the lifecycle.

When the dev picks the judge model, BizFinder ops will set `OLLAMA_KEEP_ALIVE=5m` on the existing Ollama service and provision a separate user on the GPU host for the judge process. Zero new infrastructure for the shared-Ollama path; GPU is free for the voice bot the rest of the time.

13. API Endpoint Reference

13.1 GET /api/qa/health

Cheap auth probe. Run this first.

```
curl -sS -H "X-QA-Secret: $QA_SHARED_SECRET" \
  https://bizfinder.ai/api/qa/health

# 200 OK (live verified):
{ "ok": true, "service": "qa-read-api", "serverTime": "2026-05-14T00:12:34.381Z"

# 401 if secret wrong or missing; 503 if env var unset on server
```

13.2 GET /api/qa/conversations

List voice conversations, newest first. Cursor-paginated.

PARAM	TYPE	DESCRIPTION
siteId	string	Filter to a tenant. Recommended — without it you see all voice traffic across tenants.
since	ISO-8601	Returns rows with <code>createdAt >= since</code> . Use for incremental polls.
limit	int 1-200	Default 50.
cursor	string	Pass <code>nextCursor</code> from previous response to paginate.

```
curl -sS -H "X-QA-Secret: $QA_SHARED_SECRET" \
  "https://bizfinder.ai/api/qa/conversations?siteId=qa-judge&limit=20"
```

```
# 200 OK (live verified shape):
```

```
{
  "ok": true,
  "count": 1,
  "nextCursor": "cmp3pknl1001k11wxbbuwbecw",
  "conversations": [
    {
      "id": "cmp3pknl1001k11wxbbuwbecw",
      "sessionId": "voice-1778655521124-24klmm",
      "siteId": "mk24x7.com",
      "businessName": "Mukul",
      "isVoice": true,
      "satisfaction": null,
      "score": null,
      "metrics": null,
      "createdAt": "2026-05-13T06:58:41.125Z",
      "updatedAt": "2026-05-13T06:58:41.125Z"
    }
  ]
}
```

13.3 GET /api/qa/conversations/<id-or-sessionId>

One conversation with every `Message` . Path segment accepts either Prisma `id` (cuid) or `sessionId` .

```
curl -sS -H "X-QA-Secret: $QA_SHARED_SECRET" \
https://bizfinder.ai/api/qa/conversations/voice-1778655521124-24klmm
```

200 OK (live verified shape):

```
{
  "ok": true,
  "conversation": {
    "id": "cmp3pkn11001k11wxbbuwbecw",
    "sessionId": "voice-1778655521124-24klmm",
    "siteId": "mk24x7.com",
    "businessName": "Mukul",
    "isVoice": true,
    "messages": [
      { "id": "...", "role": "assistant",
        "content": "Hello, you've reached Mukul, how can I help you?",
        "rating": null, "createdAt": "2026-05-13T06:58:41.918Z" },
      { "id": "...", "role": "user",
        "content": "How long will it take to get a website?",
        "rating": null, "createdAt": "2026-05-13T06:58:41.920Z" },
      { "id": "...", "role": "assistant",
        "content": "It typically takes around 4 to 8 weeks ...",
        "rating": null, "createdAt": "2026-05-13T06:58:41.921Z" }
    ],
    "metrics": null
  }
}
```

HTTP STATUS CODES

STATUS	MEANING
200	OK
400	Bad since (not ISO-8601)
401	Missing or wrong X-QA-Secret
404	No conversation with that id/sessionId
503	Server doesn't have QA_SHARED_SECRET set

14. Smoke Test

Save as `~/qa-smoke-test.sh` inside the account, `chmod +x` , run. Source is also in the repo at `docs/qa-smoke-test.sh` .

```
#!/bin/bash
export QA_SHARED_SECRET="fea77692e1d33bc18807e6dda9be4f95db471ba289af20ea737e500
export MAIN_HOST="https://bizfinder.ai"      # or staging.bizfinder.ai

# 1. Health probe (expect 200)
curl -fsS -H "X-QA-Secret: $QA_SHARED_SECRET" "$MAIN_HOST/api/qa/health" | jq .

# 2. List conversations (expect 200 with count + conversations[])
curl -fsS -H "X-QA-Secret: $QA_SHARED_SECRET" \
  "$MAIN_HOST/api/qa/conversations?limit=5" | jq .

# 3. Wrong secret should 401
code=$(curl -sS -o /dev/null -w "%{http_code}" -H "X-QA-Secret: wrong" "$MAIN_HOST/api/qa/health")
[ "$code" = "401" ] && echo "AUTH GATE OK" || echo "AUTH BROKEN ($code)"
```

15. Developer Scope of Work

What the third-party developer owns:

1. **Bot-side audio capture.** Extend the Pipecat FrameProcessor in bot/app.py (lines ~438–510) to buffer user PCM (UserStartedSpeakingFrame → TranscriptionFrame) and TTS PCM (TTSStartedFrame → TTSStoppedFrame). Gated by a QA flag so non-QA calls remain in-memory-only.
2. **Audio persistence.** Write per-turn WAV + full-call WAV, mp3-encode on hangup, SFTP to analytics@38.247.189.143:2203 into ~/turns/<sessionId>/<n>--<role>.mp3 and ~/recordings/<sessionId>.mp3 .
3. **Judge pipeline.** Pull call metadata via the QA API, pair with the audio files, run the judge model (e.g. Qwen2.5-Omni via Ollama with OLLAMA_KEEP_ALIVE=5m for auto-unload), produce scores. Output destination is the dev's choice.

16. Pending on BizFinder Side

ACTION	OWNER	WHEN
Create dedicated <code>qa-judge</code> siteId via admin panel	BizFinder admin	Before dev starts work
Allocate A6000 GPU slot for judge model	BizFinder ops	Once dev signals readiness for live testing
Set <code>OLLAMA_KEEP_ALIVE=5m</code> on existing Ollama service	BizFinder ops	If sharing Ollama with the voice bot
Hand <code>MODAL_START_URL</code> + <code>VOICE_PROMPT_URL</code> to the developer	BizFinder dev	At engagement start

17. Rotation & Revocation

If credentials leak or the engagement ends, BizFinder ops will:

- **Rotate `QA_SHARED_SECRET`** and redeploy the main app on prod and staging.
- **Rotate the SSH password** for the `analytics` account.
- **Disable the `analytics` account** and archive its home directory.

The developer does not need any of this access during normal use — it is on the BizFinder side. Contact BizFinder if you suspect a credential leak.